

28/05/2026

AI4SE

CORPUS FORGE

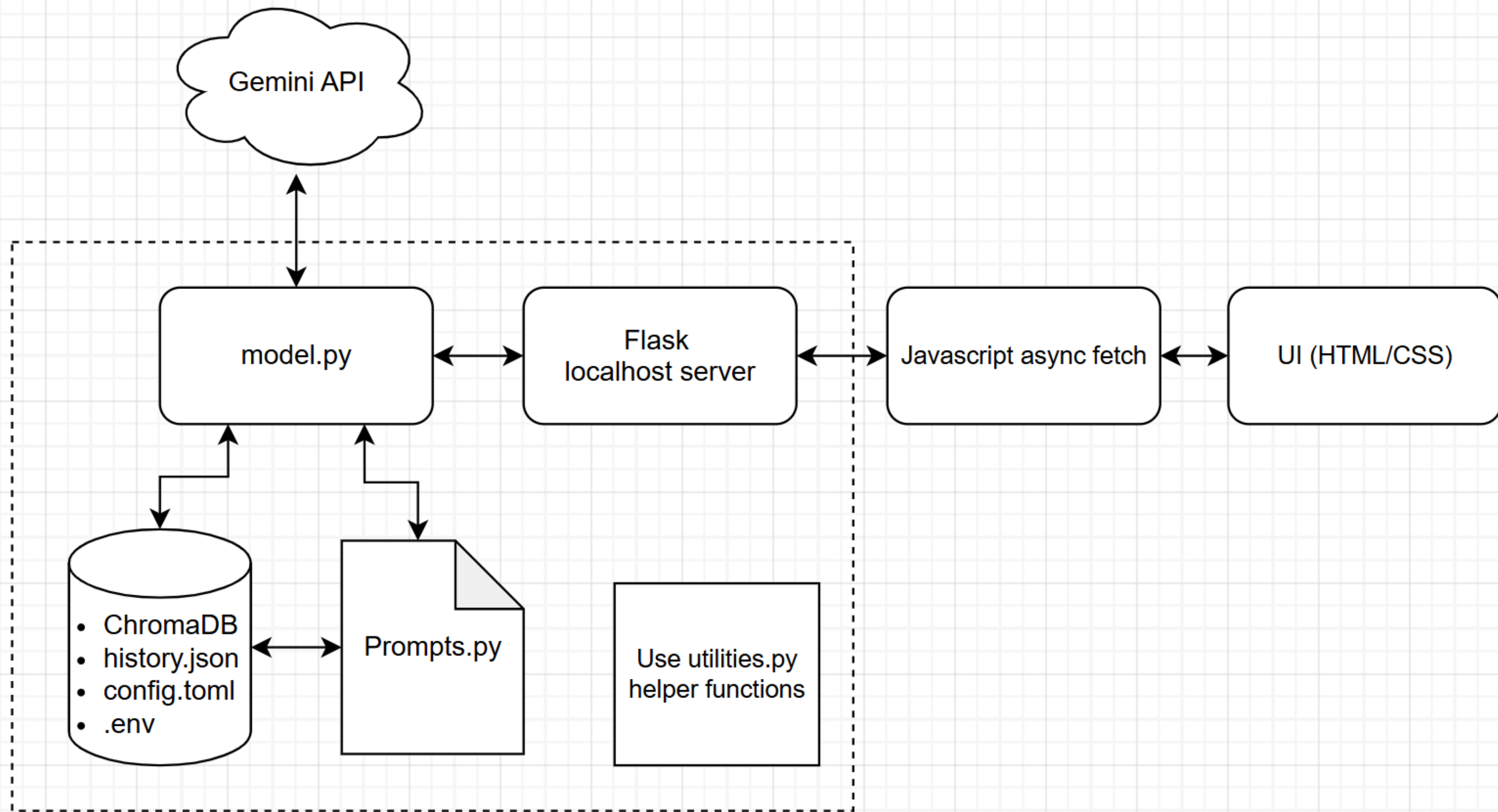
ARTEM & SUCCESS & YUCHEN

MOTIVATION

0x01

OVERVIEW

First is the overview of our
Capstone Project



OUR MOTIVATION

Si

SIMPLE

Something we can
manage and understand
ourselves

Ef

EFFICIENT

Isn't too bloated and easy
enough on the hardware

UX

PLEASEANT UX

Doesn't require too much
setup, user friendly

WHAT

0X02

WHY

EXPLAIN OUR TECH CHOICES

ARCHITECTURE

CONFIG FILES

WHAT WE HAVE

**TWO CONFIGURATION FILES
THAT SERVE DIFFERENT
PURPOSES**

.ENV

SENSITIVE SECRETS

Google API key

.TOML

**NON-SECRET APPLICATION
SETTINGS**

Values control how the AI behaves



HISTORY.JSON

app saves user and model messages there in Gemini-compatible format



OUTPUT/

for generated HTML content, such as quizzes, flashcards, or visualization pages

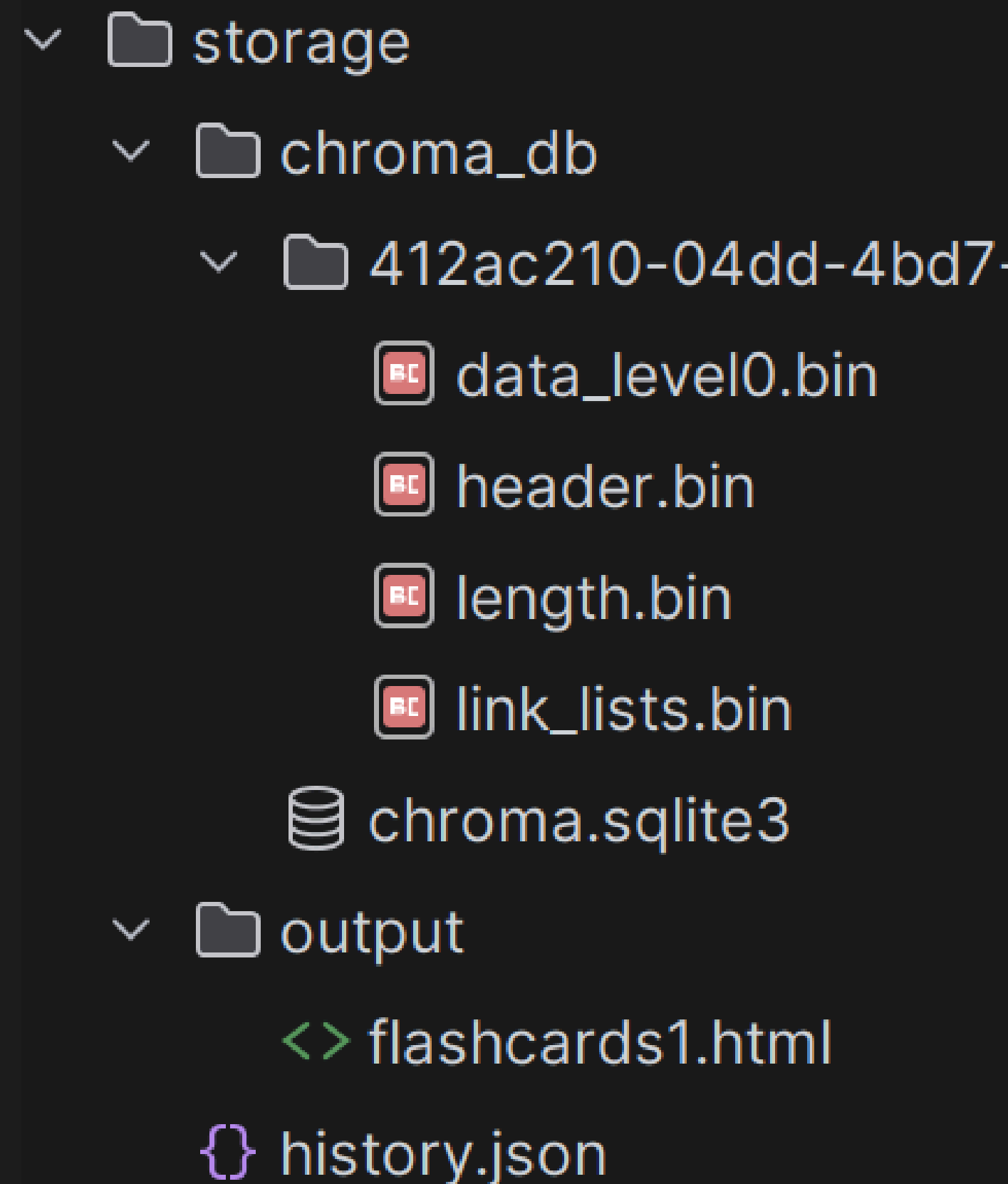


CHROMA_DB/

where parsed document chunks are saved as embeddings

USED FOR LOCAL RUNTIME DATA

STORAGE FOLDER



WHY FLASK

THREE ADVANTAGES & ONE DISADVANTAGE

LIGHTWEIGHT AND SIMPLE

Flask does not force a complex project structure, which fits this small web app well.

EASY INTEGRATION

Flask is Python-based, integration is straightforward.

GOOD FOR RAPID PROTOTYPING

Flask makes it quick to create endpoints, is useful for a capstone project.

LARGE PROJECTS REQUIRE MANUAL STRUCTURE

Flask is flexible, but it does not enforce architecture

BETTER USER EXPERIENCE

The page does not need to fully reload.

GOOD FOR AI WAITING TIME

Gemini responses may take a few seconds, so async requests let the UI stay responsive.

CLEANER FRONTEND/ BACKEND SPLIT

Flask returns JSON data, and JavaScript updates the page dynamically.

ASYNCHRONOUS JAVASCRIPT

+ PROMISES

ARCHITECTURE

OTHERS

AI RAG workflow

UNSTRUCTURED FOR PARSING FILES INTO CHUNKS (AND STORING IN CHROMADB)

LARGE FILES CANNOT BE SENT
ENTIRELY INTO THE PROMPT, AND
RETRIEVAL WORKS BETTER WITH
SMALLER, MORE PRECISE TEXT PIECES



CHROMADB FOR CALCULATING EMBEDDINGS

CHROMADB IS MAINLY USED
TO STORE AND QUERY
EMBEDDINGS



GEMINI API

GEMINI PROVIDES READY-
TO-USE LARGE LANGUAGE
MODEL CAPABILITIES

ENGINEERING CHALLENGES

OVERCOMING DIFFICULTIES

**ERROR HANDLING
(CANT INTERRUPT AI,
CANT GENERATE
WITHOUT CONTEXT,
INITIALIZING FILES,
CORRECT COLLECTION
NAMES...)**

**INSTANTIATING
GEMINI.CHAT ONLY
ONCE (AIMODEL CLASS)**

**ENABLING
PERSISTANCE**

0X03

Built the website interface
and focused on layout,
interaction, and user
experience.

**FRONTEND
DEVELOPER**

Developed the core
logic, APIs, and data
processing.

BACKEND DEVELOPER

Tested the system,
reported issues, fixed bugs,
and checked the final
functionality.

TESTER / DEBUGGER

0X04

TEAMWORK DISTRIBUTION

AI COLLABORATION

0X05 - THE POWER OF ARTIFICIAL INTELLIGENCE

SHARING

Sharing good prompt &
prompt results with each
other

ASKING

Using ask mode
(Guide AI, not the other way
around)

PLANNING

asking it to create a plan for
several people
(e.g. I'm doing backend, my
colleague is doing frontend)

01

**SEPARATING WORK
(BACKEND, FRONTEND,
DOCUMENTATION)**

02

**TECH: USING WEB APIS,
RAG CONTENT
GENERATION, ASYNC
JAVASCRIPT FOR AUTO
PAGE UPDATES**

03

**COOPERATING
WITH AI**

OUR 2 WEEKS CRASH COURSE IN FULL STACK DEV

LESSONS LEARNED

COMMUNICATION

0X07 - WORKING TOGETHER



**COMMUNICATING
ONLINE AND IN
PERSON**

MANAGING VERSIONS

Members—3

-  Artem
Visual Studio Code
-  凌九 UNRD
PyCharm Professional
-  PuLSe
Visual Studio Code

(3AM)

TECH OTAKUS SAVE THE WORLD

THANKYOU

LET'S BUILD SOMETHING THAT LASTS

Artem Iavchunovskii

Success Aderibigbe

Yuchen Zheng